

8.4.34

EE25BTECH11031 - Sai Sreevallabh

Question:

Let (x, y) , be any point on the parabola $y^2 = 4x$. Let $\mathbf{P}$ be the point that divides the line segment from $(0, 0)$ to (x, y) in the ration 1:3. Then the locus of $\mathbf{P}$ is.

- 1) $x^2 = y$ 2) $y^2 = 2x$ 3) $y^2 = x$ 4) $x^2 = 2y$

Solution

Given points can be taken to be

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ and } \mathbf{Q} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (4.1)$$

Point $\mathbf{P} = \begin{pmatrix} x_p \\ y_p \end{pmatrix}$ divides the line segment $\mathbf{P} - \mathbf{O}$ in the ratio 1:3. so

$$\mathbf{P} = \frac{3\mathbf{O} + \mathbf{Q}}{3 + 1} \quad (4.2)$$

$$\Rightarrow \begin{pmatrix} x_p \\ y_p \end{pmatrix} = \frac{1}{4} \begin{pmatrix} x \\ y \end{pmatrix} \quad (4.3)$$

So,

$$x = 4x_p \text{ and } y = 4y_p \quad (4.4)$$

Substituting the above in the equation in the parabola $y^2 = 4x$.

$$y_p^2 = x_p \quad (4.5)$$

Therefore, the locus of the point $\mathbf{P}$ is the parabola $y^2 = x$.

